

Definition 24 (Absolutely Convergent Series). *An infinite series*

$$\sum_{n=1}^{\infty} a_n$$

is said to be absolutely convergent if the series

$$\sum_{n=1}^{\infty} |a_n|$$

is convergent.

Remark 6. *If $a_n \geq 0$ for every n , then*

$$|a_n| = a_n,$$

and hence

$$\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} a_n.$$

Therefore, a series of non-negative terms is convergent if and only if it is absolutely convergent.

Definition 25 (Conditionally Convergent Series). *A series is said to be conditionally convergent if it is convergent but not absolutely convergent.*

Example 41. *Consider the constant sequence*

$$a_n = c, \quad n \in \mathbb{N},$$

where $c \in \mathbb{R}$ is fixed. Determine whether the series

$$\sum_{n=1}^{\infty} a_n$$

is convergent.

Solution. *The n th partial sum is*

$$s_n = a_1 + \cdots + a_n = nc.$$

If $c = 0$, then $s_n = 0$ for every n , and hence the series converges.

Now suppose $c \neq 0$. By the Archimedean Property,

$$|nc| \longrightarrow \infty.$$

Therefore, the sequence (s_n) is divergent, and hence the series is divergent. Thus,

$$\sum_{n=1}^{\infty} a_n$$

is convergent if and only if $c = 0$.

Example 42 (Geometric Series). Let $z \in \mathbb{R}$ satisfy $|z| < 1$. Consider the infinite series

$$\sum_{n=0}^{\infty} z^n.$$

Since the n th partial sum is

$$s_n = \sum_{k=0}^n z^k = \frac{1 - z^{n+1}}{1 - z},$$

and

$$z^{n+1} \longrightarrow 0,$$

it follows that

$$s_n \longrightarrow \frac{1}{1 - z}.$$

Hence,

$$\sum_{n=0}^{\infty} z^n = \frac{1}{1 - z}, \quad |z| < 1.$$

Example 43 (Telescoping Series). Let (a_n) and (b_n) be two sequences satisfying

$$a_n = b_{n+1} - b_n, \quad n \in \mathbb{N}.$$

Then the n th partial sum is

$$\begin{aligned} s_n &= a_1 + a_2 + \cdots + a_n \\ &= (b_2 - b_1) + (b_3 - b_2) + \cdots + (b_{n+1} - b_n) \\ &= b_{n+1} - b_1. \end{aligned}$$

Hence, the series

$$\sum_{n=1}^{\infty} a_n$$

is convergent if and only if the sequence (b_n) is convergent. In that case,

$$\sum_{n=1}^{\infty} a_n = -b_1 + \lim_{n \rightarrow \infty} b_n.$$

Example 44. Evaluate the series

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)}.$$

Solution. Observe that

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}.$$

Thus,

$$a_n = \frac{1}{n(n+1)}, \quad b_n = -\frac{1}{n}.$$

Since

$$\lim_{n \rightarrow \infty} b_n = 0,$$

the previous example gives

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = -b_1 + \lim_{n \rightarrow \infty} b_n = 1.$$

Example 45 (Harmonic Series). Recall Example 18. The harmonic series

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

is divergent.

Example 46 (p -Series). For $p > 0$, consider the series $\sum_{n=1}^{\infty} \frac{1}{n^p}$. Then

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

is convergent if and only if $p > 1$. Equivalently,

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = \begin{cases} \text{convergent}, & p > 1, \\ \text{divergent}, & 0 < p \leq 1. \end{cases}$$

In particular, the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent, whereas the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent.

Theorem 18 (Cauchy Criterion). The series $\sum_{n=1}^{\infty} a_n$ converges if and only if for

every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$n, m \geq N \implies |s_n - s_m| < \varepsilon,$$

where (s_n) denotes the sequence of partial sums.

Thus, the series $\sum_{n=1}^{\infty} a_n$ converges if and only if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$n > m \geq N \implies |a_{m+1} + a_{m+2} + \cdots + a_n| < \varepsilon.$$

Proof. The series $\sum_{n=1}^{\infty} a_n$ converges if and only if the sequence (s_n) of partial sums converges. By the Cauchy Criterion for sequences, this is equivalent to saying that (s_n) is a Cauchy sequence. Since

$$s_n - s_m = a_{m+1} + a_{m+2} + \cdots + a_n,$$

the second statement follows immediately. $\square$

Theorem 19. *If the series $\sum_{n=1}^{\infty} a_n$ converges, then $a_n \rightarrow 0$.*

Proof. Let $s_n = \sum_{k=1}^n a_k$ be the sequence of partial sums. Since the series converges, (s_n) converges. Hence, by the Cauchy Criterion for sequences, for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$|s_n - s_m| < \varepsilon, \quad \forall n, m \geq N.$$

Now let $n \geq N + 1$. Then $n - 1 \geq N$, and therefore $|a_n| = |s_n - s_{n-1}| < \varepsilon$. Hence $a_n \rightarrow 0$. $\square$

Theorem 20. *Suppose that the series $\sum_{n=1}^{\infty} a_n$ converges to s . Then, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that*

$$\left| \sum_{n=N+1}^{\infty} a_n \right| < \varepsilon.$$

Proof. Let $\varepsilon > 0$ be given. Since $s_n \rightarrow s$, there exists $N \in \mathbb{N}$ such that $|s - s_N| < \varepsilon$. Since $\sum_{n=N+1}^{\infty} a_n = s - s_N$, it follows that

$$\left| \sum_{n=N+1}^{\infty} a_n \right| = |s - s_N| < \varepsilon.$$

This completes the proof.

Note: For some $k \in \mathbb{N}$, the sum $\sum_{n=k}^{\infty} a_n$ is called the tail of the series. $\square$

Theorem 21 (Algebra of Convergent Series). *Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be two convergent series with sums A and B , respectively. Then:*

1. *The series $\sum_{n=1}^{\infty} (a_n + b_n)$ is convergent and*

$$\sum_{n=1}^{\infty} (a_n + b_n) = A + B.$$

2. *For every $\lambda \in \mathbb{R}$, the series $\sum_{n=1}^{\infty} \lambda a_n$ is convergent and*

$$\sum_{n=1}^{\infty} \lambda a_n = \lambda A.$$

Theorem 22. *If the series $\sum_{n=1}^{\infty} |a_n|$ converges, then the series $\sum_{n=1}^{\infty} a_n$ also converges.*

Proof. Let $s_n = \sum_{k=1}^n a_k$ and $\sigma_n = \sum_{k=1}^n |a_k|$ denote the sequences of partial sums of $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} |a_n|$, respectively.

Since (σ_n) converges, it is a Cauchy sequence. Thus, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$|\sigma_n - \sigma_m| < \varepsilon, \quad \forall n > m \geq N.$$

For $n > m \geq N$, we have

$$|s_n - s_m| = \left| \sum_{k=m+1}^n a_k \right| \leq \sum_{k=m+1}^n |a_k| = \sigma_n - \sigma_m < \varepsilon.$$

Hence, (s_n) is a Cauchy sequence. By the Cauchy Criterion for series, the series $\sum_{n=1}^{\infty} a_n$ converges. $\square$

3.2 Test of convergence

Theorem 23 (Comparison Test). *Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be two series of non-negative real numbers. Suppose that $a_n \leq b_n$ for all $n \in \mathbb{N}$. Then:*

1. *If $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ also converges.*
2. *If $\sum_{n=1}^{\infty} a_n$ diverges, then $\sum_{n=1}^{\infty} b_n$ also diverges.*

Example 47. The series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent. Since

$$0 \leq \frac{1}{n^3} \leq \frac{1}{n^2}, \quad \forall n \in \mathbb{N},$$

the Comparison Test implies that $\sum_{n=1}^{\infty} \frac{1}{n^3}$ is also convergent.

Example 48. The harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent. Since

$$\frac{1}{n} \leq \frac{1}{\sqrt{n}}, \quad \forall n \in \mathbb{N},$$

the Comparison Test implies that $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ is also divergent.

Theorem 24 (d'Alembert's Ratio Test). Let $\sum_{n=1}^{\infty} a_n$ be a series of positive real numbers. Suppose that

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = r.$$

Then:

1. If $0 \leq r < 1$, then the series $\sum_{n=1}^{\infty} a_n$ converges.
2. If $r > 1$, then the series $\sum_{n=1}^{\infty} a_n$ diverges.
3. If $r = 1$, then the test is inconclusive.

Example 49. Consider the series $\sum_{n=1}^{\infty} \frac{1}{n!}$. Since

$$\frac{a_{n+1}}{a_n} = \frac{1}{n+1} \longrightarrow 0,$$

the Ratio Test implies that the series converges.

Example 50. Consider the series $\sum_{n=1}^{\infty} \frac{2^n}{n}$. Since

$$\frac{a_{n+1}}{a_n} = \frac{2^{n+1}}{n+1} \cdot \frac{n}{2^n} = 2 \cdot \frac{n}{n+1} \longrightarrow 2 > 1,$$

the Ratio Test implies that the series diverges.

Example 51. Consider the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$. Since

$$\frac{a_{n+1}}{a_n} = \frac{n}{n+1} \rightarrow 1,$$

the Ratio Test is inconclusive. Indeed, the harmonic series is divergent.

On the other hand, for the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$,

$$\frac{a_{n+1}}{a_n} = \left(\frac{n}{n+1} \right)^2 \rightarrow 1,$$

yet this series is convergent. Thus, no conclusion can be drawn when the limit equals 1.

Theorem 25 (Cauchy's Root Test). Let $\sum_{n=1}^{\infty} a_n$ be a series of positive real numbers. Suppose that

$$\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = r.$$

Then:

1. If $0 \leq r < 1$, then the series $\sum_{n=1}^{\infty} a_n$ converges.
2. If $r > 1$, then the series $\sum_{n=1}^{\infty} a_n$ diverges.
3. If $r = 1$, then the test is inconclusive.

Example 52. Consider the series $\sum_{n=1}^{\infty} \frac{1}{2^n}$. Since

$$\sqrt[n]{a_n} = \sqrt[n]{\frac{1}{2^n}} = \frac{1}{2},$$

the Root Test implies that the series converges.

Example 53. Consider the series $\sum_{n=1}^{\infty} 2^n$. Since

$$\sqrt[n]{a_n} = \sqrt[n]{2^n} = 2,$$

the Root Test implies that the series diverges.

Example 54. Consider the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$. Since

$$\sqrt[n]{a_n} = \sqrt[n]{\frac{1}{n}} = \frac{1}{n^{1/n}} \rightarrow 1,$$

the Root Test is inconclusive. Indeed, the harmonic series is divergent.

On the other hand, for the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$,

$$\sqrt[n]{a_n} = \sqrt[n]{\frac{1}{n^2}} = \frac{1}{n^{2/n}} \rightarrow 1,$$

yet this series is convergent. Thus, no conclusion can be drawn when the limit equals 1.

Example 55. Show that the series $\sum_{n=1}^{\infty} \frac{2^n n!}{n^n}$ converges.

Solution. Let $a_n = \frac{2^n n!}{n^n}$. Then

$$\frac{a_{n+1}}{a_n} = \frac{2^{n+1}(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{2^n n!} = 2 \left(\frac{n}{n+1} \right)^n.$$

Since $(1 + \frac{1}{n})^n \rightarrow e$, we have

$$\left(\frac{n}{n+1} \right)^n = \frac{1}{(1 + \frac{1}{n})^n} \rightarrow \frac{1}{e}.$$

Hence

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \frac{2}{e} < 1.$$

Therefore, by the Ratio Test, the series $\sum_{n=1}^{\infty} \frac{2^n n!}{n^n}$ converges.

Example 56. Evaluate the following limits:

$$1. \lim_{n \rightarrow \infty} \frac{(n!)^{1/n}}{n}.$$

$$2. \lim_{n \rightarrow \infty} \frac{n!}{n^n}.$$

Brief Solution.

$$1. \text{ Taking logarithms, } \log \left(\frac{(n!)^{1/n}}{n} \right) = \frac{1}{n} \sum_{k=1}^n \log \left(\frac{k}{n} \right). \text{ This is a Riemann sum for}$$

$\int_0^1 \log x \, dx = -1$. Hence

$$\lim_{n \rightarrow \infty} \frac{(n!)^{1/n}}{n} = \frac{1}{e}.$$

2. We have $0 \leq \frac{n!}{n^n} = \prod_{k=1}^n \frac{k}{n} \leq \left(\frac{1}{2}\right)^{\lfloor n/2 \rfloor}$. Since the right-hand side tends to 0, the Sandwich Theorem gives

$$\lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0.$$

Example 57. Determine whether the series $\sum_{n=1}^{\infty} \frac{7^{n+1}}{9^n}$ is convergent.

Solution. Let $a_n = \frac{7^{n+1}}{9^n}$. Then

$$\frac{a_{n+1}}{a_n} = \frac{7^{n+2}}{9^{n+1}} \cdot \frac{9^n}{7^{n+1}} = \frac{7}{9}.$$

Since

$$\frac{7}{9} < 1,$$

the Ratio Test implies that the series $\sum_{n=1}^{\infty} \frac{7^{n+1}}{9^n}$ converges.

Theorem 26 (Abel-Pringsheim Criterion). Let (a_n) be a decreasing sequence of non-negative real numbers. If the series $\sum_{n=1}^{\infty} a_n$ converges, then $na_n \rightarrow 0$.

Proof. Since $\sum a_n$ is convergent, there exists N such that $|s_n - s_m| < \varepsilon$ for $n, m \geq N$. For $k \geq N$,

$$ka_{2k} \leq a_{k+1} + \cdots + a_{2k} = s_{2k} - s_k.$$

Thus,

$$\lim_{n \rightarrow \infty} (2n)a_{2n} = 0.$$

Since $a_{2n+1} \leq a_{2n}$, we have

$$(2n+1)a_{2n+1} \leq (2n+1)a_{2n} \leq (2n)a_{2n} + a_{2n}.$$

Now the series $\sum a_n$ is convergent, so the sequence (a_n) converges to 0, and hence the subsequence (a_{2n}) converges to zero. It follows from above that the sequence $((2n+1)a_{2n+1})$ is convergent. Using Exercise 15, we conclude that $na_n \rightarrow 0$. $\square$

Consider the series

$$\sum_{n=2}^{\infty} \frac{1}{n \log n}.$$

Although

$$\lim_{n \rightarrow \infty} \frac{n}{n \log n} = 0,$$

the series nevertheless diverges. We will see later, using the integral test, why this series diverges.

Example 58. Show that the series $\sum_{n=1}^{\infty} \frac{1}{an+b}$, where $a, b > 0$, is divergent.

Solution. Let $a_n = \frac{1}{an+b}$. Then (a_n) is a decreasing sequence of non-negative real numbers. If the series $\sum_{n=1}^{\infty} a_n$ were convergent, then by the Abel–Pringsheim Criterion (26), we would have $na_n \rightarrow 0$.

However,

$$na_n = \frac{n}{an+b} = \frac{1}{a + \frac{b}{n}} \rightarrow \frac{1}{a} \neq 0.$$

This contradiction shows that the series $\sum_{n=1}^{\infty} \frac{1}{an+b}$ is divergent.

Exercise 17 (Abel’s Test). Assume that $\sum a_n$ is convergent and (b_n) is either increasing or decreasing, and there exist $m, M \in \mathbb{R}$ such that

$$m \leq b_n \leq M, \quad \forall n \in \mathbb{N}.$$

Then $\sum a_n b_n$ is convergent.

Definition 26. A series of the form

$$\sum_{n=1}^{\infty} (-1)^{n-1} a_n$$

or

$$\sum_{n=1}^{\infty} (-1)^n a_n,$$

where $a_n \geq 0$ for all $n \in \mathbb{N}$, is called an alternating series.

Example 59. The following are alternating series:

1. $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$.
2. $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} = -\frac{1}{1} + \frac{1}{4} - \frac{1}{9} + \frac{1}{16} - \cdots$.
3. $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2^n} = \frac{1}{2} - \frac{1}{4} + \frac{1}{8} - \frac{1}{16} + \cdots$.

Theorem 27 (Leibniz Test). *Let (a_n) be a decreasing sequence of non-negative real numbers such that $a_n \rightarrow 0$. Then the alternating series*

$$\sum_{n=1}^{\infty} (-1)^{n-1} a_n$$

converges. Moreover, if

$$S = \sum_{n=1}^{\infty} (-1)^{n-1} a_n,$$

then

$$a_1 - a_2 \leq S \leq a_1.$$

Proof. Let

$$s_n = \sum_{k=1}^n (-1)^{k-1} a_k.$$

Observe that

$$s_{2n} = (a_1 - a_2) + \cdots + (a_{2n-1} - a_{2n}).$$

Since $a_{2n+1} \geq a_{2n+2}$,

$$s_{2n+2} - s_{2n} = a_{2n+1} - a_{2n+2} \geq 0.$$

Hence (s_{2n}) is increasing. Moreover,

$$s_{2n} = a_1 - (a_2 - a_3) - \cdots - (a_{2n-2} - a_{2n-1}) - a_{2n} \leq a_1.$$

Thus (s_{2n}) is bounded above, and therefore converges. Let

$$s_{2n} \longrightarrow S.$$

Now,

$$s_{2n+1} = s_{2n} + a_{2n+1}.$$

Since $a_{2n+1} \rightarrow 0$, we obtain

$$s_{2n+1} \longrightarrow S.$$

Hence, both the even and odd subsequences of (s_n) converge to the same limit. Therefore,

$$s_n \longrightarrow S,$$

which proves that the series converges.

Finally,

$$a_1 - a_2 = s_2 \leq s_{2n} \leq S \leq s_{2n+1} \leq s_1 = a_1,$$

and letting $n \rightarrow \infty$ gives

$$a_1 - a_2 \leq S \leq a_1.$$

□

Example 60. Consider the alternating harmonic series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}.$$

Since the sequence $a_n = \frac{1}{n}$ is decreasing and satisfies $a_n \rightarrow 0$, the Leibniz Test (27) implies that the series converges.

However, the series is not absolutely convergent, because

$$\sum_{n=1}^{\infty} \left| \frac{(-1)^{n-1}}{n} \right| = \sum_{n=1}^{\infty} \frac{1}{n},$$

which is the harmonic series and is divergent (see Example 18).

Thus, the alternating harmonic series is conditionally convergent.

Example 61. Show that

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} e^{-1/n}$$

is convergent.

Solution. Let $a_n = \frac{(-1)^{n+1}}{n}$ and $b_n = e^{-1/n}$. The series $\sum a_n$ is convergent by the Leibniz Test (27).

Moreover, (b_n) is increasing and

$$0 < b_n < 1, \quad \forall n \in \mathbb{N}.$$

Thus (b_n) is bounded and monotone. Hence, by Abel's Test (17),

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} e^{-1/n}$$

is convergent.